- Measured by escaped defects.

They decided to come up with a formula that combined all of these data points into a single metric that would indicate the overall health of releases across their development teams, a code health determination (CHD). If this metric started to trend in the wrong direction, they could put the brakes on and dig into the problem. If the metric stayed in a good place, then teams could continue developing and deploying at will.

In their metric each of the four elements would make up 25% of the total value, and each metric would have equal weight in the total calculation. If F represents a formula that normalized each metric, a high level of the formula would be represented like this:

```
F1(CLOC) + F2(Estimate Health) + F3(Recidivism) + F4(Escaped
Defects)
```

They decided to create a final number that was a score between 0 and 100 where 0 meant everything was going to pieces and 100 meant everything was perfect. To create this number they had to do some math to normalize the inputs and the outputs.

Normalizing changed lines of code

After exploring their data the team concluded that each small change they made was around 50 CLOC. To come to their indicator they decided to divide CLOC by 50, chop off the decimal, and then multiply it to magnify the effect. Because 25 would be the best possible value, they subtracted the result from 25. To handle the potential of a negative number, they took the maximum between the result and 0 to ensure they would stay within the bounds of 0–25. This gave them the following function:

```
MAX((25 - ABS((int)(cloc/50)-1)*5), 0)
```

With that formula, example inputs and outputs are shown in table 7.2.

Table 7.2. Example inputs and outputs after normalizing the ideal number of CLOC